

Track H, phase 1 in the garage: assessment, material and the week

Companion sheet to lab/PROTOCOL_H4.md, lab/LAB_GUIDE_H4.md and
PREREGISTRATION_13.md

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2026-10-08 Preprint v1.2: doi:10.5281/zenodo.23228685

1. Where the work stands (an honest reading)

The preprint *Logarithmic boundary depth and the conditioning of the discrete inverse conductance problem on hyperbolic lattices* (v1.2, 18 pages) establishes, with every computation preregistered and every refutation kept:

- On layer-truncated hyperbolic tilings the condition number κ of the Dirichlet-to-Neumann sensitivity Jacobian grows polynomially in N ($\log_{10} \kappa = 3.89$ at $N = 847$ on $\{7, 3\}$); on square and triangular disks it grows as $e^{c\sqrt{N}}$ (ball arithmetic: 16.0, 16.5, 27.0 decades at $N = 421, 797, 1069$). All tested networks are exactly full rank over two finite fields: the difference is conditioning, not identifiability.
- Across five hyperbolic tilings $\log \kappa$ is nearly a function of maximal depth alone (spread ≤ 0.56 decades at equal depth while N varies $15\times$); flat lattices are ≥ 1.8 decades worse at equal depth.
- The mechanism is measured: the per-edge boundary signal decays alike on all lattices, but the boundary signatures of equal-depth edges stay nearly independent on hyperbolic tilings (effective dimension fraction ≥ 0.79 at every depth) and collapse on flat disks (0.14–0.39); the depth-restricted κ grows by a lattice constant per unit depth on flat disks (1.1–1.2 decades square, 1.7–1.8 triangular) against 0.3–0.4 on hyperbolic ones. A harmonic-measure band-counting argument predicts the flat collapse ($d \cdot f_d$ constant within $1.26\times$ on instances it had not seen) but not yet the constant.
- In simulation a single-node defect is localised at the 3×10^{-4} precision budget on both geometries; the hyperbolic advantage is a noise margin ($\geq 100\times$ at $N \approx 316$); component tolerance to 5%, offsets to 10%, gain drift to 1% and a 10-bit ADC do not break localisation at $N \approx 112$.

Assessment. A solid, modest, well-documented contribution to discrete inverse problems; not a landmark. Its strongest feature is the method (preregistration, a claim ledger with evidence digests, refutations in the open), which reviewers respect. Its limits are real: at matched probe count the advantage is at most half a decade; the flat growth constant is unexplained; there is no theorem; one external review so far. It says nothing about holography or AdS/CFT, and must not be presented as if it did. Possible relevance: sensor layouts and tomography on designed graphs.

2. Why the garage test is a good first experiment

One number, fixed before the first solder joint: the ratio of relaxation times of a physical $\{7, 3\}$ $L=2$ board ($N = 112$, 140 resistors, 35 capacitors) and a square $R=6$ board ($N = 113$, 200 resistors, 69 capacitors), both built from the same component batches and calibrated against a one-resistor, one-capacitor cell from those batches.

Prediction	Statement	Band (preregistration 13)
T1	$\tau(\{7, 3\} L=2) / \tau(\text{square } R=6)$	0.604 ± 0.03
T2	τ / τ_{cal} for $\{7, 3\} L=2$ and square $R=6$	2.750 ± 0.08 and 4.551 ± 0.14
S1	six boundary-pair resistances per board (DMM)	all within 3% of the predicted ohms

Nominal $RC = 0.100$ s ($R = 100$ k Ω , $C = 1$ μ F): $\tau = 0.275$ s and 0.455 s. The test is cheap and hard to fool, and a refutation with all gates passing would be informative (the board model, not the mathematics, would be wrong). What this phase cannot test is the conditioning advantage itself: that is phase 2 (a multiplexed 77-contact map), for which the simulations say the chain bought now is sufficient.

3. Material for the week

Components (two boards, two calibration cells)

Item	Qty	Specification and note
Resistor 100 k Ω	352	metal film 1%, one batch : 140 + 200 + 2 calibration + 10 spare
Capacitor 1 μ F	112	film (PET/PP) or C0G, one batch: 35 + 69 + 2 + 6 spare; <i>no</i> class-2 ceramic (DC bias), <i>no</i> electrolytic (leakage)
Perfboard or stripboard	2	$\geq 15 \times 15$ cm, one per board (or a printed jig following the schematic)
Bus wire, tinned copper	3 m	boundary rail and ground bus; plus hookup wire in two colours
Quad op-amp, rail-to-rail input	1	MCP6004 / TLV2374 class, with a 14-pin socket; unity-gain buffers: 1 driver + 3 probes
ADC module ADS1115	1	16 bit, 860 SPS; the ESP32 internal ADC (10-bit effective) is excluded (+20 to +80% τ error on the virtual bench)
Microcontroller	1	ESP32 (or Raspberry Pi Pico) with USB cable: step on a GPIO, I ² C readout, timestamps
Decoupling	6	4×100 nF ceramic at the op-amp and ADC supply pins, 2×10 μ F on the rails (not in the generated BOM)
Probe pads, terminals	8	6 header pins or pads (nodes 8, 7, 0 and 10, 20, 31), 2 screw terminals (rail, ground)

Tools and bench

- Digital multimeter, 4.5 digits preferred (sorting at 2%, static pairs at 3%).
- Soldering iron with a fine tip, solder, flux, desoldering wick, tip cleaner; flush cutters, needle-nose pliers, tweezers, a board holder (“third hand”).
- 3.3 V supply for the step (bench supply or the ESP32 regulator); USB power for the logger.
- Laptop with Python 3 and the Arduino IDE, the repository checked out; room thermometer and hygrometer (logged every session).
- Ventilation or a fume extractor, safety glasses. Voltages are 3.3–5 V DC: the hazards are the iron and the fumes.

Printed before session 0 (`python3 lab/make_forms.py`)

- both schematics, `lab/out/hyp73_L2_schematic.pdf` and `square_R6_schematic.pdf`;
- both solder-order tables, `lab/out/hyp73_L2_wiring.md` and `square_R6_wiring.md`;
- `lab/forms/components_form.csv` (342 resistors, 106 capacitors, one line each);
- `lab/forms/static_pairs_form.csv` (12 boundary pairs with their predicted resistance);
- `lab/forms/session_log_template.md` (one copy per session).

4. The week, gate by gate

Session	Work	Gate before the next session
0 (laptop, 1 h)	run the four lab scripts (netlist, virtual bench, analysis demo, forms); print the forms	the demo prints both verdicts True ; nothing bought before this
1 (2 h)	buy; measure every part with the DMM into the component form; sort into piles; keep 2 R + 2 C for the cells	G1: R within 2%, C within 10%
2 (4 h)	build {7,3} $L=2$ per schematic and wiring table, deepest class first, DMM after each row; probe pads 8, 7, 0; calibration cell	G2: ≥ 5 of 6 boundary pairs within 3% (rail disconnected)
3 (5 h)	build square $R=6$ likewise; probe pads 10, 20, 31	G2 on the square board
4 (3 h)	followers, ADS1115, ESP32, flash <code>step_logger.ino</code> ; five records on the calibration cell alone	G4: τ_{cal} within 5% of 0.100 s, records within 1%
5 (2 h)	five step records per board, same temperature; fill <code>h4_records.json</code> ; run the analysis; do not open the predictions	G3: three probes agree on τ within 3%
6 (1 h)	claim H4-X-0001, <code>ledger_add.py</code> , <code>experiment_gate.py</code> , results block in <code>LAB_RESULTS.md</code> , commit	ALL PASS; any verdict is reported with equal prominence

A gate that fails stops the sequence: diagnose (guide §6: τ too long means an open resistor or a missing capacitor's neighbour; too short means leakage or a class-2 ceramic; probes disagreeing means a buffer or a wrong pad), fix, re-run the gate. Numbers go from `h4_step_response.json` into the results file and the ledger; none is typed from memory.